Articulation Skill: A Patient-Navigator Regime for Public-Good Mental-Health AI

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Preprint v1.1, 2026-05-04. License: CC BY 4.0. Companion paper: *Engineering-Ethics Scaffolding for Deployed Public-Good AI Mental-Health Systems* (Paper A), Zenodo DOI 10.5281/zenodo.19941457. Code and reference implementation: https://github.com/wudaming00/stay.

Conflict of interest (front-loaded per AIES 2026 position-track norms): the author developed and operates the reference implementation discussed in §5 and has no commercial revenue from it (the code is MIT-licensed, the safety-critical prompt sections are restricted-use-licensed per Paper A, and the deployment is operated free of charge to users). No employment relationship with any named comparator system or model provider. The author intends to seek non-commercial donor and grant funding for continued operation. Full disclosure in §8.1.

Cite as: Wu, D. (2026). *Articulation Skill: A Patient-Navigator Regime for Public-Good Mental-Health AI*. Zenodo. DOI 10.5281/zenodo.20028720.

Abstract

Between an undifferentiated, distressed user and the professional help they may eventually receive lies a gap of two kinds: *time* (often weeks of waiting, or earlier, the user not yet knowing they need help at all) and *information* (the user not yet knowing what kind of help to ask for). The five existing AI mental-health regimes (closed-clinical, open-companion, crisis line, symptom checker, directory) do not occupy this between-position: each presupposes the user already knows what kind of system they have walked into. This paper proposes a sixth regime sited explicitly in the gap, *articulation skill*: an AI that helps the user translate undifferentiated distress into the vocabulary referral systems index on, without recommending a provider and without rendering a clinical diagnosis. The regime is operationalized by three required properties (patient-owned system-agnostic output; bounded non-DSM vocabulary spanning modality, credential, level-of-care, specialty, and phenomenological labels; refusal of provider recommendation) plus a scope-recognition constraint for out-of-scope populations. A 2024–2026 prior-art search finds seven closest neighbors, none shipping the conjunction. The paper proposes an *anti-engagement criterion* category, operationalized by handoff rate paired with a phased appropriateness audit, as a structural counter to the engagement metrics dominating the AI-companion industry. Lineage engages five literatures (Strauss/Star, Mishler/Kleinman, Nutbeam/Hibbard, Elwyn/decision-aids, AI-safety specification-gaming canon); full disambiguation in supplementary Appendix B. Stay (thestay.app, open-source) is the author's partial reference implementation, described at architectural level only; behavioral evidence is deferred to IRB-reviewed future work, with an external clinician audit committed by 2026-06-30 under Wu 2026's deployment-pause posture. The contribution is the regime definition and the anti-engagement criterion category; the operationalization and the implementation are research-program proposals, not validated instruments.

1 · Introduction

1.1 The between-position and the bifurcation framing

The structural position this paper proposes for AI in mental-health care is *between* the undifferentiated distressed user and the professional help they may eventually receive. The two gaps the position fills are familiar to anyone who has tried to find a therapist while distressed: a *time gap* (weeks before an appointment, or earlier, not yet knowing one is needed) and an *information gap* (the user does not yet know what to ask for, what kind of clinician handles what, what modalities exist for what they are experiencing). The five existing AI mental-health regimes (closed-clinical, open-companion, crisis line, symptom checker, directory) each presuppose the user already knows what kind of system they have walked into; none is positioned in the gap.

By late 2025, the deployed AI mental-health landscape was conventionally framed as a bifurcation: closed commercial clinical systems on one side, open-source companion and roleplay systems on the other, with crisis lines, AI symptom checkers, and therapist-matching directories sometimes named as adjacent regimes. §2 enumerates these five regimes with representative deployments and property tuples.

This paper cuts the landscape on three product-level dimensions — provider recommendation, system coupling, and vocabulary scope — chosen as the minimum dimension set under which the articulation function becomes visible as a separate slot. Existing analytic frameworks cut on different dimensions (Torous et al. on passive-vs-active data collection in digital phenotyping; Olawade et al.'s 2024 AI-in-mental-health review on deployment context; NICE EVA HTG756 on funnel role for one care system; the stepped-care literature on patient-state intensity tier), under each of which articulation-skill systems land in the same bucket as one or more of the regimes named here. The five-regime construction that follows is a consequence of the dimension choice and is offered for clarity, not as a claim to authoritative classification. The known methodological hazard — a taxonomy designed to surface a function tends to surface it — is mitigated by independence at the product level: §3.1–§3.3 give per-product checks that do not require accepting the five-regime cut, and §4's prior-art finding (no current production system ships the three-property conjunction) survives reframing under any alternative taxonomy whose dimensions admit the same checks.

What matters for this paper's claim is that no existing regime occupies the between-position above, and that the regime proposed in §3 does not collapse onto any of the five by reframing.

That this user need actually surfaces on general-purpose LLMs in non-trivial volume is supported by OpenAI's own October 2025 disclosure (Strengthening ChatGPT's Responses in Sensitive Conversations, 2025-10-27) of suicidal-planning-indicator flag rates among weekly active users. The figure is industry-self-reported, classifier-confounded, and counted at message-event level rather than enrolled-user level; it is cited as a directional signal that the unmet conversation volume on general LLMs is non-zero on a tool whose operators caveat that it is not built for the use case, not as a quantified population estimate. §7.5.4 returns to the in-scope-population sizing question.

1.2 The function this regime performs (defined by what it does, not who it

serves)

A position-by-user-state framing fails almost immediately. The same user can oscillate between "thinking out loud about a hard week" and "naming an active-suicide-ideation moment" within a single session. We define the regime by the function it performs (articulation work), not by the state of the user it serves (which oscillates). The articulation function is in scope whenever a user describes experience that they do not yet know how to translate into the vocabulary referral systems index on. The function sits upstream of crisis intervention (the system must still escalate appropriately) and upstream of therapy (the system must still hand off to professional care).

Operationally, occupying this between-position requires the AI to perform several roles simultaneously for the user: knowing the user (listening with attention to phenomenological detail), teaching the user (about themselves and about the psychology vocabulary that referral systems index on), providing information when needed (modality, credential, level-of-care), making situation-dependent judgments about whether and what kind of professional help to direct toward, and recognizing crisis moments and routing accordingly. The regime defined in §3 is the structural specification of these roles, and the §5 reference implementation operationalizes them via six system-prompt agency functions.

Population estimates for the addressable need are difficult to source directly. The National Survey on Drug Use and Health (NSDUH) 2023 reports 49.5 million U.S. adults with any mental illness in the past year, of whom approximately half received no treatment. Among reported reasons for not receiving treatment, the largest single category historically is "thought I could handle the problem without treatment"; "did not know where to go for services" appears as a documented but smaller category alongside cost, stigma, time, and insurance navigation difficulty. We do not claim the articulation gap is the dominant barrier; we claim it is one of several, plausibly addressable for 10–25% of the untreated population, and not currently served by any of the five named regimes.

1.3 The thesis of this paper

The thesis is that the between-position named in §1.1, where the user is filling time-and-information gaps before reaching a clinician, defines a regime in its own right: *articulation skill*. The regime is distinct from the five regimes named in §2, currently unclaimed in published literature and product positioning, and operationalized by three required properties (§3). Seven closest neighbors each diverge on at least one of the three (§4); an open-source partial reference implementation exists, described at the architectural level (§5). A success-metric *category* is proposed in §6, an *anti-engagement criterion*, as a structural counter-indicator to the engagement metrics of the AI-companion regime.

The primary contribution is the joint construction of two coupled elements: an *articulation skill* regime in mental-health AI (§3, defined by three required properties, a scope-recognition constraint, and a deployment-side defense-of-record requirement) and an *anti-engagement criterion* category (§6, defined by anti-engagement-under-adversary and welfare-monotone properties) that the regime requires to be coherent. The category is portable to any public-good consumer AI; the regime is its canonical operationalization within the mental-health AI vertical, and the regime cannot be measured with engagement metrics without violating its own constraints. The §4 / §A.2 prior-art finding (no current production system ships the three-property conjunction) is a supporting empirical observation; §3 lead-in and §7.5.3 read it as a

structural prediction about commercial incentive, not a calendar-fact gap-claim. Handoff rate is offered as one operationalization of the category, awaiting Phase-1 validation, not as a validated metric. The engineering-ethics governance scaffolding (Wu 2026, summarized in §5) is cited as the operational frame within which the reference implementation runs and is not itself a contribution of this paper. The paper does not present empirical evaluation of efficacy. Anticipated objections are engaged in §7.5.

1.4 Intellectual lineage in one paragraph (full disambiguation in Appendix B)

The regime draws on five literatures whose canonical claims *partially support* and *partially cut against* the design choices in §3:

- **Star (1991) category-work and the CSCW articulation-work tradition** (Strauss 1985, 1988, 1993; Schmidt & Bannon 1992): the §3.3 vocabulary-bounding move places the regime on the *standardization* side of the Schmidt-Bannon distinction, while Star's category-work tradition predicts the §7.4 population-harm list as the structural output. The term *articulation skill* is repurposed from a CSCW concept whose canonical meaning is cooperative-coordination work; we keep the term despite the inversion (defense in supplementary §B.1) because of its colloquial clinical/intake usage and because Schmidt-Bannon's prediction about downstream cooperative articulation work is one of the regime's load-bearing acceptances.
- **Mishler (1984) voice-of-medicine and Kleinman explanatory models**: the regime performs upstream the biomedical standardization Mishler critiques and Kleinman asks the clinician to resist; we accept this cost on the grounds that the patient's gain in navigation legibility outweighs lifeworld content lost in pre-articulation, and the existing alternative (the user routes to ChatGPT) loses the same content with worse consequences.
- **Nutbeam 2000 health literacy and Hibbard et al. 2004 PAM**: the regime operates in Nutbeam's interactive-literacy band; PAM is a longitudinal trait-level scale, not a session-level state, and within-session articulation outcomes are not directly comparable to PAM stage transitions.
- **Elwyn et al. 2012 shared-decision-making and the patient-decision-aids tradition** (Stacey et al. 2024 Cochrane CD001431; IPDAS): the regime is kin to but not a sub-type of decision aids; three structural differences hold the distinction, detailed in §B.4.
- **AI safety / specification-gaming canon** (Krakovna et al. 2020; Goodhart 1975 and Manheim & Garrabrant 2018; Hubinger et al. 2019; Sharma et al. 2023; Casper et al. 2023; Shah et al. 2022; Perez et al. 2022): engaged substantively in §6.2 and §6.4.

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2 · The five existing regimes

For each existing regime, this section summarizes the framing, names representative deployments, identifies the user need it serves, and gives the regime's tuple against the three properties this paper uses to define the proposed sixth regime (defined in §3): provider recommendation (does the system output specific clinicians?), system coupling (is the user output portable to other care systems, or coupled to one?), and vocabulary scope (modality/credential/level-of-care literacy, DSM-diagnostic categories, both, or neither).

2.1 Closed-clinical regime (Therabot, Wysa Health, Limbic Access, Lyra

Polaris, IESO)

Care delivered by an AI agent under internal clinical review (and, for some, regulatory pathway). Need served: *direct AI-delivered therapeutic intervention*. Representative: Therabot (Heinz et al. 2025 NEJM AI RCT); Wysa Health (FDA Breakthrough Device, NHS-integrated); Limbic Access (UKCA Class IIa, Habicht et al. 2024 *Nature Medicine*); Lyra Polaris ("clinical-grade," 2025-10-14, employer-benefit network); IESO Digital Program (NHS, 2024 non-inferiority study).

2.2 Open-companion regime (Replika, Character.AI, open-source mental-health LLMs)

Conversational presence offered as emotional companion, sometimes with roleplay framing; safety sections often strippable on open-source fork. Need served: *parasocial connection / emotional presence*. The fork-strips-safety failure mode is documented (NEDA Tessa, May 2023; generative replacement of a scripted helpline produced eating-disorder-promoting advice within hours). Garcia v. Character Technologies (M.D. Fla. 6:24-cv-01903; teen-suicide tort cluster settled 2026-01-07) is the first major U.S. tort case cluster associated with this regime.

2.3 Crisis-line regime (988, Crisis Text Line, DV hotline)

Trained human voice or text in real time; authority to dispatch police and enforce protocols (including, in many states, mandatory reporting). Need served: *imminent-danger intervention*. Does not address non-crisis presentations or modality navigation.

2.4 AI symptom-checker regime (Babylon, Ada, K Health, UCSD self-triage prototype)

Conversational AI applying medical decision protocols to triage between self-care, primary care, and emergency care. Need served: *general-medical triage*. Mental-health presentations addressed at coarse granularity only (e.g., "see a doctor about anxiety"); modality, credential, and level-of-care literacy are out of scope.

2.5 Therapist-matching directory regime (Psychology Today, Headway, Alma, ZocDoc, WithTherapy)

Searchable clinician databases, often with insurance verification. Need served: *finding a specific provider given that you know what to look for*. The user-side prerequisite (knowing what kind of provider, modality, specialty, credential) is assumed. WithTherapy adds an AI matching layer downstream of the directory; the matching does not produce user-portable literacy.

2.6 Comparison table

Regime	Provider rec	System coupling	Vocabulary scope	Output to user
Closed-clinical (§2.1)	Within own network	Coupled	Modality + diagnostic	The therapy itself

Open-companion (§2.2)	None	Agnostic	None (undifferentiated)	Conversation/ relationship
Crisis line (§2.3)	None (human resource itself)	Single acuity	None	Direct intervention
AI symptom checker (§2.4)	Tier-level routing	Mostly agnostic	Loose diagnostic	Care-tier suggestion
Matching directory (§2.5)	Yes (primary)	Coupled to directory	Provider attributes	List of clinicians
Articulation skill (§3)	**None (by design)**	**Agnostic**	**Modality + credential + LOC + phenomenological labels; NOT DSM categories**	**Patient-owned literacy**

The articulation regime is the only row in which all four columns simultaneously hold the values shown.

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3 · The sixth regime: articulation skill

Definition. *Articulation skill* is the capacity of an AI to translate undifferentiated distress into the vocabulary that referral systems index on, without recommending a specific provider and without rendering a clinical diagnosis.

The regime is defined operationally by three required properties (§3.1–§3.3) plus a scope-recognition constraint (§3.4) defining out-of-scope populations. These three properties are the three product-level dimensions named in §1.1 as the minimum set under which the articulation function surfaces as a separate slot; §4.5's collapse mapping is supplementary evidence that each property does load-bearing work in the conjunction, not the primary defense for the dimension choice. Whether "currently unclaimed" is a calendar fact or a structural property is engaged in §7.5.3: closest-neighbor players have business arrangements that create substantial switching cost from system-coupled to patient-portable output, so the unclaim is best read as a structural prediction about commercial incentive, not a temporal accident.

3.1 Property 1 — Patient-owned, system-agnostic output

The artifact the user takes away from an interaction is literacy in a specific bounded vocabulary (defined in §3.3): a paraphrase of their experience in modality terms, a level-of-care suggestion phrased as a question the user can ask, named credential categories, named phenomenological labels for what the user described. The artifact lives with the user and is portable. Nothing about it presupposes the user will go to a particular network, app, directory, or care system. The user can take their articulated need to their primary care physician, to their existing therapist, to Psychology Today, to Headway, to their insurance member-services line, to their EAP, or to a friend who works in the field. The "any door" framing has a constraint: the doors are open for the in-scope population (§3.4); for the out-of-scope strata named in §7.4, the constraint is structural per the §3.3 vocabulary-bounding choice (Appendix B.1) and not an edge case to be patched.

What is the concrete artifact? In the reference implementation, the literacy persists in three places: (a) within the conversation transcript itself, which the user can export from local storage; (b) in the user's "kept insights," starred sentences the user marks during the conversation, persisted in IndexedDB; (c) optionally, in a DBT-style structured log if the user opts in. The deliberate non-design choice is that there is no single "articulation-summary PDF" the system generates as an output object. The artifact is the user's own record of the conversation, with vocabulary they can re-read and re-deploy. The position-paper consequence: a regime claim built on "patient-owned literacy" should specify the artifact, and the specification here is intentionally open (the conversation log + opt-in structured journal). A specimen redacted artifact will be deposited as supplementary material on Zenodo. The choice to not standardize a single artifact format is itself part of property 1: the system does not prescribe the form the patient's literacy takes.

This property is what makes the regime the user's tool, not a system's intake funnel. It is the property that distinguishes the regime from Limbic Access (which feeds NHS Talking Therapies as a single specific service), from Wysa Gateway (which feeds a partner provider's EHR), from Spring Health Guide (which routes within the Spring Health network), from Lyra Polaris (which routes within the Lyra network), from any matching directory, and from the human "digital navigator" role (Lancet Digital Health, 2022, which by definition operates inside one health system).

The patient-side ownership consequence (the user controls the artifact and decides where to deploy it) is the regime-level property; the operator-side rationale (avoiding the liability and quality-control responsibility that come with provider recommendation) is a derived consequence of the property, not a justification of it. The patient's epistemic outcome — leaving the conversation knowing what kind of help to ask for, at any door — is the reason the property is regime-defining.

3.2 Property 2 — Provider recommendation refused by design

The system does not output specific clinician names, does not interface with directory APIs, does not make appointment offers, does not maintain awareness of the user's insurance network, and does not recommend a specific care system or program by name. The refusal is by design. The reasoning, downstream from property 1, is that the moment an AI begins outputting specific provider recommendations, the artifact stops being patient-owned literacy and becomes a referral the user is implicitly asked to act on. The system has now positioned itself as a quality-controlling intermediary between the user and a specific provider, with two consequences: (a) the user's downstream verification of fit, credential, insurance, and availability is short-circuited by an AI's recommendation that did not include any of those checks; (b) the operator absorbs liability and curation responsibility for the directory it is in effect maintaining.

The articulation regime hands off the "what to look for" to the user and lets the user perform the "who to see" step in whichever existing referral system they prefer. The existing referral systems handle the directory work competently and have the verification infrastructure (insurance check, license verification, scheduling) the AI does not.

Refusal is not free. A user who asks "I think I need a trauma specialist, who should I see?" and receives a refusal may experience the response as rebuff and route to a general-purpose LLM, which will sometimes confabulate a name (§4.4(ii)). The forgone-benefit cost (the user obtaining a worse alternative because the regime declined to give them a worse-but-still-imperfect direct

answer) is real, and the regime accepts it on the grounds that the existing referral systems are the right venue for the "who to see" step and that an AI's confabulation rate is a worse failure mode than the friction of pointing the user to a directory. Property 2 is the design choice that the friction is the better trade.

3.3 Property 3 — Vocabulary scope: a three-tier distinction

The vocabulary the system uses to articulate the user's need is bounded by an explicit three-tier distinction.

Tier (a) is in scope: phenomenological labels and care-navigation vocabulary. The system may name experiences the user described using common phenomenological vocabulary clinicians use at intake (descriptive labels for what happened, not diagnostic categories), and may name modality, credential, level-of-care, and specialty. The vocabulary spans five sub-categories (phenomenological labels; modality across CBT-family / trauma / relational-dynamic / mindfulness / format axes; credential; level-of-care across MH and SUD continua; specialty); the full table with representative items, the per-term U.S. inclusion criterion (gates: U.S. patient-facing referral directories, recognized credentialing or training organizations, CMS HCPCS/CPT or ASAM Criteria), the country-agnostic form of the criterion (jurisdiction-specific gateway / credentialing-body / payor-taxonomy substitution), and notes on phenomenological-vs-DSM-episode disambiguation, candidate-modality deferrals, and credential disambiguation are in §A.3. The regime category travels across jurisdictions; the vocabulary list does not.

Tier (b) is out of scope: DSM diagnostic categories. The system does not say "this sounds like generalized anxiety disorder," "you may have PTSD," "this is consistent with major depressive episode," "this is borderline personality disorder," "you have ADHD." Diagnostic statements are clinical acts with downstream consequences (insurance coding, treatment planning, self-concept, credentialing impacts) that an AI without the patient's history, formal examination, and clinical context is not positioned to perform without harm. The literature on diagnostic-disclosure effects on self-concept (Yanos, Roe, & Lysaker 2010 on illness identity and self-stigma; Lally 1989 on the engulfment construct) is consistent with this design choice. Naming a phenomenological label (panic attack, tier a) and naming a diagnostic category (panic disorder, tier b) are different clinical acts; the regime allows the first and refuses the second.

The productive move when a user asks a tier-(b) question (e.g., "am I bipolar?") is described here so the regime is defined by what it *does* in addition to what it refuses. The system declines to give a yes/no diagnostic answer, names that the question is one a clinical evaluation can answer, describes what such an evaluation would assess (sleep architecture changes, episode duration, family history, functional change), names which credentials perform such evaluations (psychiatrist, PMHNP, psychologist with diagnostic-assessment training), and offers phenomenological vocabulary the user can deploy in describing the experience to that evaluator (elevated mood, decreased need for sleep, racing thoughts, impulsive spending). The user leaves with literacy enabling them to *get* the diagnostic question answered, not with the diagnostic answer itself. This is the regime-coherent productive move and §6.2's appropriateness audit rates whether it was performed.

Tier (c) is in scope as triage signal but out of scope as diagnosis: acuity recognition. When a user presentation crosses a threshold for acute risk (suicidal ideation with method or plan or intent per C-SSRS levels 3–5 — *C-SSRS* is the Columbia Suicide Severity Rating Scale (scale © 2008 Research Foundation for Mental Hygiene; validation in Posner et al. 2011 *Am J Psychiatry*

168(12):1266–1277); reports of voices commanding harm to self or others; active substance impairment; current intimate-partner violence with physical harm or strangulation history), the system must recognize the threshold and surface appropriate human resources. The system describes the risk indicator phenomenologically ("voices telling you to hurt someone is a kind of experience that needs an in-person evaluation now") and does not name the diagnostic-grade subtype ("command hallucination") whose forensic implications require formal assessment. This recognition is acuity-routing, not diagnosis; it does not require naming a disorder; it requires naming the risk and the resource. This is the boundary the §5 reference implementation operates under.

The boundary between (a) and (b) is operationally fuzzy in practice (is "this sounds like the kind of thing CBT is built for" implicitly diagnostic of an anxiety presentation?). The guardrail the reference implementation uses: modality is named as a candidate frame the user can ask about, never as a fit-claim about the user's condition. "DBT was developed for emotion-regulation difficulty; that is a thing some people find this kind of presentation responds to" is allowed; "you have BPD and need DBT" is not.

This three-tier distinction differentiates the regime from AI symptom checkers (which produce diagnostic-shaped outputs even when phrased cautiously) and from any AI claiming clinical interpretation. It is also a sharper line than "no symptom-to-diagnosis" alone: it is symptom-to-vocabulary, with the vocabulary explicitly bounded and the diagnostic line explicitly named.

3.4 Scope-recognition constraint: out-of-scope presentations

The three properties above describe what the regime does. A scope-recognition constraint, derived from the population-harm analysis in §7.4, defines what the regime must not attempt on its own: presentations for which articulation by the AI is the wrong intervention.

The out-of-scope list is not a binary; clinically it is a stratified judgment. The relevant strata:

- *Psychotic-spectrum presentations*, in three sub-tiers. (i) Florid active psychosis, mania with psychotic features, acute attenuated-psychosis-syndrome / CHR-P: out of scope; community mental health and CSC programs are the destination. (ii) Stabilized-on-medication schizoaffective, psychotic depression in remission, brief psychotic disorder in resolution: *partially in scope* for navigation around medication management and outpatient continuity, with explicit deference to the user's prescriber. (iii) Prodromal and emerging psychosis (circumstantial speech, ideas-of-reference vs delusions of reference, excessive coherence-seeking): the failure-mode shape the regime is structurally bad at recognizing, since LLMs are biased to reward articulate, coherence-seeking interlocutors. §4.4(iv) engages this.
- *Severe cognitive impairment, advanced dementia, TBI*: out of scope; geriatric MH and PCP-coordinated pathways are the destination.
- *Low-literacy and non-English-first users*: vocabulary literacy assumes a baseline of education and language fluency that excludes substantial populations.
- *Non-Western somatized presentations*: cultural variations (somatic depression, family-mediated explanatory frameworks per Kleinman; *nervios / ataque de nervios* in Latino populations; South Asian shame-honor frameworks) can be mis-articulated by a Western-modality vocabulary.
- *Minors under unsupervised conditions*: the reference implementation has an 18+ self-attest age gate (IRB-aligned, not state-minor-consent-statute-aligned; some states allow 12+ to

consent to outpatient MH services without parent involvement; mature-minor doctrine applies in some jurisdictions). Adolescents 12–17 in minor-consent jurisdictions are out of scope for this implementation; future work could address with parent/guardian framing.

- *Suicide presentations stratified by C-SSRS level and prior-attempt status*: naming "active SI without plan" collapses multiple C-SSRS ideation levels (Posner et al. 2011: level 1 wish to be dead, 2 non-specific active thoughts, 3 thoughts with methods, 4 intent without plan, 5 ideation with plan and intent). Prior attempt is the strongest empirical predictor of completed suicide (Bostwick et al. 2016). The imminent-risk SOP (LICENSE-PROMPT §1.a) covers C-SSRS 4–5 and any prior-attempt-positive presentation; C-SSRS 1–2 are typically in scope for articulation with safety-planning handoff (Stanley & Brown 2012); C-SSRS 3 is borderline, where the deployment-pause posture and SOP overlap (the C-SSRS-level mapping is this paper's clinical interpretation; Wu 2026 §2.1 describes §1.a in terms of method-driven persuasion plus means restriction plus companion-during-call without committing to specific thresholds). Means restriction (Means Matter line) is for firearms and medications; *not* for substance-as-coping (which is MI / harm-reduction territory).
- *Eating-disorder presentations with medical instability*: below-threshold BMI, electrolyte abnormalities, bradycardia, syncope are medical emergencies requiring ED / inpatient triage, not articulation.
- *Acute substance impairment*: a transcript produced while intoxicated has compromised authorship; §8.4 data-ethics covers consent for citation but not articulation-while-impaired.
- *Acute imminent crisis*: 988 and emergency services, not articulation.

For these strata a confidently articulated wrong modality recommendation is worse than no articulation, because the user arrives at the wrong door with conviction.

The regime requires the system to recognize when an interaction is out of scope and to signal toward human navigation instead of attempting the work. Operationally this is a *scope-recognition constraint* alongside the three positive properties above: an articulation system that cannot identify presentations for which articulation is contraindicated is incompletely defined under this regime.

At the regime-definition level, scope-recognition is operationalized as classification gates fired on each user turn against a documented stratum list, with the stratum-to-action mapping (route-to-988, route-to-DV-hotline, route-to-PCP, route-to-emergency-services, deployment-pause-on-stratum-X) published as part of the regime instance's specification. The *gates* are detection logic; the *mapping* is published action policy. The reference implementation operationalizes this via the imminent-risk SOP (LICENSE-PROMPT §1.a, described in Wu 2026 §2.1) for the acute-crisis stratum and via stratum-specific protocols for the other strata where implemented. A regime instance must publish (a) the stratum list, (b) the detection mechanism for each, and (c) the mapped action; gates without published action policy do not satisfy the constraint.

False-negative cost. The scope-recognition constraint matters because its failure mode is the population-harm scenario in §7.4: when the system performs articulation on a presentation that should have been routed out, the user receives confident wrong vocabulary. The constraint is part of the regime definition; the reference implementation as of 2026-05-02 partially implements it (the imminent-risk SOP and deployment-pause posture cover the acute-crisis case; SMI, low-literacy, and cross-cultural cases remain documented gaps per §7.4). The §5 framing is therefore "implements three properties and partially implements the scope-recognition constraint," and the §5 clinician-of-record audit is the next checkpoint at which the SMI / low-

literacy / cross-cultural gaps either close or trigger deployment-pause.

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4 · Distinguishing from closest neighbors

The closest neighbors to the proposed regime were identified by a structured search across academic literature, position papers, and product positioning announcements (2024–2026; search methodology in §A.1 Appendix). Inclusion criterion for "closest neighbor": a deployment that satisfies at least one of P1–P3 unambiguously while diverging on at least one of the remaining two. The remainder of the §A.2 inventory either diverges on all three (and is treated at regime level in §2) or has a property tuple unverifiable from public positioning. We address the seven closest neighbors and a generalized eighth (general-purpose LLMs).

4.1 Seven closest neighbors — divergence summary

Neighbor	Source	P1 (patient-owned output)	P2 (refuses provider rec)	P3 (bounded non-DSM)
Limbic Access	Habicht et al. *Nat Med* 2024; Rollwage et al. *JMIR AI* 2023	✗ system-coupled (NHS Talking Therapies referral)	✗ recommends Talking Therapies tier	✗ classifies across 8 conditions (∼94%)
Wysa Gateway (US)	Press release 2025-06-11	✗ inside provider's funnel	✗ routes to integrated provider	partial
Spring Health Guide	US launch 2026-04	✗ Spring Health network-coupled	likely ✗ in-network routing	unclear
Lyra Polaris	Lyra Health, announced 2025-10-14	✗ in Lyra benefit network	partial	✗ delivers care
IESO Digital Program	2024 RCT, NHS partnership	✗ NHS partnership	✓	✗ delivers CBT
WithTherapy + similar matching	by definition	partial	✗ recommends providers	✓
Digital navigator concept	*Lancet Digital Health* 2022	(human role, not AI)	(human role)	(human role)

Limbic Access is the closest published neighbor with peer-reviewed access-and-waiting-time evidence. Spring Health Guide is the most prominent 2026 launch; if Guide's user output turns out portable and provider-rec-refusing, the regime distinction would collapse to network-coupling alone — §7.5.3 engages this class of refutation directly. IESO's "literacy plus skills" takeaway sits adjacent to the §5 *skill teacher* agency function, but differs on P3 (delivers therapy vs names candidate modalities). The digital-navigator concept is the closest conceptual precedent and differs on agency: the navigator is a human team role; the regime is the AI direct-to-patient. Matching services answer "which provider," whereas the regime answers the upstream "what kind of provider am I looking for," so the regime sits upstream of matching services rather than competing with them.

4.2 Headspace Ebb (most material defensibility risk)

Ebb launched October 2024 as a motivational-interviewing companion in the Headspace app, with a January 2026 stratified-care upgrade adding smart triage, care plans, and provider matching across 15,000+ providers. Reported scale: 7M+ messages, 4 countries (Headspace press release 2025-05-21). Ebb differs on all three properties (in-app + system-coupled to Headspace network; explicit personalized provider matching post-Jan-2026; motivational-interviewing-plus-symptom-evaluation blends care delivery with navigation). Where it is closest is *reach*, at consumer scale within the closed-network stratified-care pattern. As with Spring Health Guide, a future decoupling from the network would collapse the regime distinction to system-coupling alone; §7.5.3 engages this falsifier class.

4.3 Other industry and regulatory neighbors

A 2024–2026 sweep of ~25 patient-facing AI products (additional names: SonderMind AI Care Concierge, Lyra AI Care Search, Talkspace TALK, Jimini Health Sage, UnitedHealthcare Avery, ChatGPT Health; full inventory in §A.2). Structural finding: roughly 12 are closed-clinical, ~8 are stratified-care network-coupled navigators, ~3 are general-medical, and **zero ship the three-property combination as their design center**. NICE Early Value Assessment HTG756 (2024) on "digital front door technologies for NHS Talking Therapies" is the closest active UK regulatory vocabulary; it formally assesses Limbic Access (UKCA Class IIa, ~40% of NHS Talking Therapies services) and Wysa Digital Referral Assistant (UKCA Class I, ~31 services). The disambiguation: a *digital front door* funnels users into one care system; *articulation skill* produces patient-portable literacy for any door. Selfapy, deprexis (German DiGAs, prescription-pathway guided self-help / CBT) and Mon Sherpa (Qare, France, reportedly no longer in service) are EU closed-clinical-with-prescription-pathway neighbors confirming the dominant European deployment shape.

4.4 ChatGPT, Claude, and general-purpose LLMs

General-purpose LLMs are the de facto venue for the user this regime serves; operator disclaimers notwithstanding, users go there because there is nowhere else to go. Four concrete failure modes distinguish what general-purpose LLMs do in this use from what an articulation-regime system does by design.

(i) Sycophancy at long context: general LLMs are trained to agree with the user's framing, reinforcing wrong paths when the framing is wrong. Sharma et al. (arXiv:2310.13548, 2023) document the mechanism across five frontier model families: sycophancy is *trained in* by RLHF preference signals, increases with model scale, and persists despite explicit instructions otherwise. It is a base-model property, not a deployment-time problem. The regime's defenses split at two layers. *Diagnostic-claim sycophancy* ("you may have GAD") is addressed by Property 3's diagnostic refusal. *Framing sycophancy* ("you're right, this sounds like a flashback") is *not* addressed by Property 3: phenomenological labeling is permitted, and a sycophantic base model under Property 3 will produce labels agreeing with the user's framing rather than probing it. The reference implementation inherits this bias from Claude Sonnet 4.5. Alleged harm pathway: the Raine v. OpenAI complaint (San Francisco Superior Court, filed 2025-08-26) alleges 377 OpenAI-moderation-flagged messages in the conversation history of Adam Raine (16, California) prior to his April 2025 suicide; on 2025-11-06 a coalition of plaintiffs' firms (Social Media Victims Law Center; Tech Justice Law Project) filed seven additional California state-court suits,

four for decedents' families (Shamblin, Lacey, Enneking, Ceccanti) and three for AI-psychosis injury survivors (Irwin, Madden, Brooks). These are unadjudicated as of 2026-05-02; population-scale evidence rests on Sharma et al. and OpenAI's October 2025 operator-disclosed moderation-flag rate, with the Raine cluster providing alleged but not adjudicated harm pathway in specific cases.

(ii) Confabulated provider names and contacts: general LLMs in long mental-health threads occasionally produce specific therapist names, contact information, and "I called and confirmed" claims that are entirely fabricated, with high downstream-harm potential. The articulation regime forecloses this failure mode by design (Property 2: provider recommendation refused).

(iii) No safety-section retention on derivative wrappers: system prompts wrapping general LLMs for mental-health use can be silently stripped on fork, the failure mode the NEDA Tessa case made vivid in 2023. The articulation regime addresses this through the governance pattern in Wu 2026 (LICENSE-PROMPT, deployment-pause posture, rule-coverage CI).

(iv) AI-induced or AI-amplified psychosis: documented in 2024–2025 case-series literature (Sakata 12-patient series, *Psychiatric News* 2025-10; *JMIR Mental Health* AI-delusional-state case reports 2025); OpenAI's October 2025 disclosure noted ~0.07% weekly-active-users showing psychosis/mania indicators. Two mechanism layers. *Deployment-layer mechanisms* (confirmatory engagement with delusional content; coherent phenomenological labeling of misinterpreted experiences that entrenches a delusion with clinical-sounding language) are partially defended by Property 3 plus §3.4 scope-recognition. *Base-model-layer mechanisms* (long-context coherence reinforcement that the user mistakes for confirmation-of-reality; sycophantic validation of user-presented framing) are not addressable by Property 3 / §3.4 alone; the regime therefore requires every conformant deployment to specify a deployment-side defense-of-record against this layer (per §6.2: turn-budget cap, runtime accommodation-pattern monitoring, anti-sycophancy fine-tuning, or base-model substitution). Prodromal and emerging psychosis (circumstantial speech, ideas-of-reference, excessive coherence-seeking) is the register an LLM is biased to reward as engaged conversation; recognition-threshold scope-recognition catches the florid case and misses the prodromal case (the modal harm pathway). The reference implementation's specific defense-of-record selection and adequacy is reviewed at the §5 clinician audit.

4.5 Collapse paths (§3 promise, delivered)

§3 claims that relaxing any one of the three properties collapses the regime onto an existing one. The mapping:

Property relaxed	Collapse target
P1 patient-owned output → system-coupled	Closed-clinical (§2.1) or stratified-care navigator (Limbic, Wysa Gateway, Spring Health Guide, Lyra Polaris, Headspace Ebb post-Jan-2026, Wysa DRA)
P2 provider recommendation refused → recommendation produced	Therapist-matching directory regime (§2.5; WithTherapy, Headway, Alma)
P3 vocabulary scope bounded → diagnostic categories produced	AI symptom-checker regime (§2.4; Babylon, Ada, K Health) or closed-clinical

P3 vocabulary scope bounded → undifferentiated conversation	Open-companion regime (§2.2; Replika, Character.AI, general-purpose LLMs in default mental-health use)

Two of the three properties relaxed simultaneously typically collapses to a single existing regime; all three relaxed is a generic mental-health AI without distinguishing posture.

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5 · Reference implementation — Stay

Stay (thestay.app, MIT-licensed code at github.com/wudaming00/stay, restricted-use license on safety-critical prompt sections) is the author's open-source reference implementation. It is cited as architectural existence proof for the §3 three-property conjunction; behavioral evidence is deferred to §10.3. Stay implements §3.1–§3.3 and partially implements the §3.4 scope-recognition constraint (acute-crisis case covered via an imminent-risk SOP; SMI, low-literacy, and cross-cultural cases remain documented gaps per §7.4). Six system-prompt agency functions perform the articulation work: *translator* (paraphrasing user experience into modality / credential / level-of-care vocabulary); *mirror* (reflecting back what the user said with phenomenological precision so they hear themselves clearly); *navigator* (situation-dependent direction toward appropriate human resources, including the explicit judgment that no professional help is needed when that is the right answer); *bridge-with-companion-during-call* (supporting the user before, during, and after a crisis-line call); *skill teacher* (teaching psychology vocabulary and self-understanding the user can re-deploy across future encounters); *logger* (DBT-style structured recording, opt-in). The system runs under a published engineering-ethics governance pattern (Wu 2026, Zenodo DOI 10.5281/zenodo.19941457): a restricted-use system-prompt license (LICENSE-PROMPT, with clauses for an imminent-risk SOP, a leverage-prevention rule, a third-party-characterization restriction, and a companion-during-call requirement designated preserved-pending-clinical-validation), a deployment-pause posture with an external clinician-of-record audit committed by 2026-06-30 as the substantive trigger, and a heartbeat dead-man-switch returning referral-only on operator-unavailability beyond 7 days. Session storage is local-only, end-to-end encrypted with a browser-held key, 90-day default retention; telemetry is metadata-only.

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6 · An anti-engagement criterion as success metric

The conceptual contribution of this section is a *category* of success metric: an *anti-engagement criterion*. A regime defined in opposition to the engagement-optimized AI-companion industry must propose a success metric whose increase corresponds to less user engagement with the AI itself. This contradicts the structure of the metrics that dominate the AI-companion industry (DAU, session length, retention, return rate, messages-per-day) and is the contribution the articulation regime makes to the broader question of how to measure success in public-good consumer AI. Two necessary properties define the category. *Property C1 (anti-engagement under adversary)*: a perfect-AI-companion deployment (one that maximizes session length, retention, and user dependency) should *minimize* the metric, not maximize it. This excludes engagement metrics that have been relabeled. *Property C2 (welfare-monotone)*: the metric must be empirically anchored to a downstream user-welfare outcome (clinical handoff received, articulation-quality rating by clinician, etc.), so that pathological maximizers (e.g., "user does not return for 30 days because the experience was harmful" trivially satisfies C1 but is welfare-negative) are excluded

by failing C2. C1 alone is a sign-flip on a one-dimensional metric and admits known-harmful counterexamples; C1 paired with C2 defines the category. C1 and C2 are necessary but not sufficient: distinguishing welfare-aligned disengagement (the user leaves because they got what they came for) from frustrated exit (the user leaves because the experience failed) is operationalization-layer work, addressed by §6.1's tri-valued outcome rather than tightened at the category level. The §6.2 audit checks both axes at the rubric layer: appropriateness rates the welfare-monotone direction; gaming-mode analysis rates the anti-engagement direction.

This section proposes one operationalization of the category, *handoff rate*, paired with a *handoff-appropriateness audit* without which the metric cannot serve as a primary outcome. The category is the citable contribution; the operationalization is offered as a research-program proposal awaiting Phase-1 validation (§6.3), not as a validated instrument.

6.1 Handoff rate (operationalization)

A session is counted as a *handoff* if (a) the system surfaced one or more named human resources during the session and (b) the user's behavior in the session's later half moves toward that resource via an unprompted intention statement ("I'll call them," "I'll text my sister," "I'll bring this to my therapist Thursday").

Three operational specifications. *Session boundary*: a session is a contiguous span of conversation within one IndexedDB session-id; multi-day articulation work across browser sessions is rolled up at the session-id level, with conversation-thread-level aggregation available for sensitivity analysis. *Within-session position*: "later half" is computed as the second half of turns within the session, since conversation length is highly variable and any fixed-fraction final-third can range from 1 turn to 30. *Unprompted detection*: the §6.1 telemetry is content-free at the population layer, so "unprompted" is not directly detectable from population telemetry and is rated only at the §6.2 appropriateness-audit layer where rater inspection of full transcripts is permitted under audit consent. Population-layer handoff rate is the *count* metric (resource-surfaced × intention-statement-emitted, both telemetry events); the *quality* of the intention (prompted vs unprompted, fitted vs premature) is rated only on the audit sample and is what makes the audit pair-with-rate non-optional. The population-layer rate alone is therefore gameable on the unprompted dimension; §6.2's audit closes that gap. The metric is the rate-and-audit pair, not the rate alone.

A no-conversion outcome is tracked separately: sessions in which the user reaches an articulation outcome (named a modality, named a level of care, named a specialty to ask about) without a within-session handoff intention. The tri-valued session outcome is therefore: handoff / served-without-handoff / disengaged-frustrated (the third identified by short-session early-termination patterns plus explicit negative-feedback signals; we acknowledge this third category is the loosest of the three at the population-telemetry layer and its accuracy is part of what the §6.2 audit rates).

A population-bias caveat. Handoff rate is self-report-anchored: a user who cannot safely emit a handoff intention (most importantly the coercive-control IPV user on a monitored device, §6.2(b) and §7.4) is structurally invisible to the metric and is recorded as *served-without-handoff* or *disengaged-frustrated*, both reading as routine in the tri-valued outcome. The metric systematically under-counts the population with the most severe risk profile. Closing the gap requires a *non-handoff* audit pathway sampled with over-representation of short-session abrupt-termination patterns on IPV-flagged sessions, rating whether absence of handoff is regime

success or regime invisibility. Until that pathway operates, the metric is structurally blind to the IPV stratum.

Denominator. Handoff rate is computed over *handoff-eligible* sessions, defined as sessions where either (a) the presentation crossed an acuity threshold under the imminent-risk SOP (LICENSE-PROMPT §1.a), or (b) the user explicitly asked about a human resource. Sessions ineligible for handoff (low-acuity articulation work that resolves to served-without-handoff appropriately) are excluded from the denominator. This closes a third gaming mode (denominator inflation): a system optimizing handoff rate over *all* sessions has incentive to push low-acuity articulation work toward acuity-threshold framings, converting served sessions into handoff sessions inappropriately. Computing the rate over handoff-eligible sessions only removes that incentive. The eligibility flag is set by the same acuity-recognition logic that drives §3.4 scope-recognition; eligibility-flag accuracy is itself part of what the §6.2 appropriateness audit rates.

Handoff rate operationalizes the anti-engagement category in two ways. It is anti-engagement by construction: a high handoff rate means users are leaving the AI for human resources, and a perfect AI-companion product (by the metrics common to that industry) would have a handoff rate of zero. It is patient-aligned rather than platform-aligned: the metric does not reward the AI for being preferred over human help; it rewards the AI for making human help reachable.

Three construct-validity limits qualify the metric. First, it is a process indicator rather than an outcome: it tracks the within-session transition from "experience without vocabulary" to "named target with intention to deploy," not downstream contact or clinical benefit, which the §3.1 patient-portable / local-only architecture deliberately does not measure (§10.1 names outcome measurement as a separate research program). Second, it is not an articulation-quality measure: two systems with the same handoff rate can produce articulation outputs of different fidelity, and the §6.2 appropriateness audit only partially addresses this on handoff sessions, leaving served-without-handoff sessions unmeasured (§10.3 names articulation quality as the separate research program). Third, intent statements under distress are weak behavioral predictors: the crisis-intervention literature (Stanley & Brown 2012 Safety Planning Intervention; Drew 2001 no-suicide-contract; Jobes 2016 CAMS) is consistent that intent statements made in acute distress unreliably predict post-session contact. Partial patches (the appropriateness audit's elicitation-quality dimension can downgrade weak intents; §10.1 names downstream-contact validation) do not convert handoff rate into a reliable behavioral predictor; the metric tracks a process state, not behavior.

6.2 Handoff-appropriateness audit (without which §6.1 is gameable and iatrogenic)

Handoff rate is gameable in three ways. Denominator inflation is addressed by computing the rate over handoff-eligible sessions only (§6.1 above). The other two operate on the numerator side and are addressed by the appropriateness audit.

The first gaming mode is premature surfacing. A handoff-maximizing system can surface 988 / DV hotline / "call your therapist" inappropriately early in a session to bank the metric. The clinical concerns with premature surfacing, in order:

(a) Iatrogenic alarm response and dissociation in trauma populations. A trauma survivor mid-articulation who hits an unsolicited "would you like to call 988" prompt experiences a re-traumatization-shaped event: the system has unilaterally decided they are in danger, which is

the abuser-pattern shape. Hyperarousal or dissociation responses follow. This is the trauma-informed-care framework's primary concern with unsolicited resource-surfacing.

(b) IPV-specific endangerment if the device is monitored. Premature surfacing of DV hotlines, escape-plan vocabulary, or shelter information on a shared or surveilled device can directly increase risk to the user. This is the most clinically severe premature-surfacing concern for the IPV population.

(c) Erosion of calibration trust. If a non-crisis user receives a crisis resource, the user (and downstream clinicians who hear about it) lose calibration trust in the system's risk-detection. This affects future risk-detection accuracy.

(d) Help-seeking-stigma pathway: premature crisis-resource surfacing for non-crisis users may teach users that AI conversation equals "you are in crisis," producing avoidance of future help-seeking. Consistent with adjacent literature (Henderson, Evans-Lacko, and Thornicroft 2013 on stigma as a barrier to help-seeking; Schomerus et al. 2012 *Acta Psychiatrica Scandinavica* meta-analysis on persistent stigma despite biogenetic-causation campaigns; Clement et al. 2015 *Psychological Medicine* systematic review of stigma and help-seeking) but not directly tested on AI-mediated premature surfacing; flagged as a hypothesis extrapolated from adjacent literature.

The defense across all four pathways is a handoff-appropriateness sub-metric, computed as a clinician-rated audit on a sample of handoff sessions, scoring whether the resource surfaced was correctly fitted to the user's presentation. Raw handoff rate without an appropriateness audit cannot serve as a primary outcome regardless of which of (a)–(d) is the dominant harm.

The second gaming mode is intention-elicitation. A system prompt could be tuned to elicit verbal "I'll call them" intentions cheaply ("would you call them?" exploits yes-bias). The defense is to count handoff only when the resource was surfaced because the presentation required it (acuity threshold met or user explicitly asked) and the intention statement was unprompted by a leading question. Operationally this requires the appropriateness audit to rate elicitation quality, not just resource fit.

Two additional caveats for any deployment of these metrics. (i) Handoff is a within-session self-reported signal, not a downstream clinical outcome; it does not measure whether the user actually contacted the resource or whether the resource helped. (ii) Handoff rate alone, without the appropriateness audit, can drive the regime toward the failure modes (1) and (2) it was meant to defend against; the pair must be reported together or neither should be reported.

Goodhart engagement and scope. The §6.2 gaming-mode analysis is *adversarial reward-shaping* (proposing a metric and asking which games it enables) rather than *reward-gaming* on observed deployed metrics (Krakovna et al. 2020). Goodhart 1975 applies ("when a measure becomes a target, it ceases to be a good measure"; taxonomized in Manheim & Garrabrant 2018, arXiv:1803.04585), and the three gaming modes named above are *adversarial Goodhart* variants. The §6.3 appropriateness audit creates a recursive Goodhart risk (once the rubric is published, it becomes the new optimization target); the available defenses (periodic rubric revision, blind raters across deployments, stratified red-teaming of the rubric) are research programs the regime needs to specify, not treat as solved. Shah et al. 2022 (arXiv:2210.01790) predicts that systems optimized against handoff rate develop the misgeneralized goal of "elicit handoff intentions" rather than "produce articulation that helps users"; the §6.2 gaming defenses address symptoms of this misgeneralization rather than its cause. The only durable address is direct

articulation-quality measurement (§10.3). The metric is proposed as an outer-alignment *evaluation target* for system-prompt-level optimization against a frozen base model; using handoff rate as an RLHF reward signal would face additional inner-alignment risks (Hubinger et al. 2019, arXiv:1906.01820: mesa-objective divergence in deployment), out of scope until that work is done.

Inheritance from the base-model substrate. The deployment inherits its base model's RLHF-induced behaviors (annotator homogeneity, distributional shift, polysemy of helpful/harmless in distress contexts, reward hacking; Casper et al. 2023). Capability vs alignment is conflated in the §6.2 audit: a system failing resource-fit may be mis-aligned or capability-limited, and the audit cannot distinguish without a separate capability-evaluation pass closed-clinical regimes have done and the articulation regime has not. The base-model-layer gap (§4.4(iv)) is the regime's open requirement: a regime-conformant deployment must specify at least one deployment-side mitigation as a defense-of-record (turn-budget cap on long sessions in IPV/SI/SMI tiers; runtime accommodation-pattern monitoring; anti-sycophancy fine-tuning; or base-model substitution); operational treatment is committed to the §5 audit checkpoint.

6.3 Phased audit specification

The audit is itself a two-phase commitment: a formative phase that stabilizes the rubric on a small unstratified sample with two licensed raters and no κ target, and a stratified phase that achieves a power-adequate κ confidence interval through deliberate over-sampling of low-base-rate, high-stakes strata (IPV, acute SI, mania, psychosis). Reporting rule: handoff rate and appropriateness scores are released as a pair, never separately, and Phase 1 is never reported as if it were Phase 2. Per-stratum sample sizes and rater logistics are deferred to a methods companion. The position-paper claim here is that a regime whose success metric requires clinician-rated audit cannot be evaluated by a single operator alone, and that the cost of getting evaluation right is part of the regime specification, not an implementation detail; §10.1 names a research-collaboration funding partner as a precondition for Phase 2.

6.4 Relation to existing evaluation instruments

The handoff-rate / appropriateness-audit pair is upstream of and orthogonal to two existing measurement traditions. VERA-MH (Belli, Bentley, et al., arXiv 2510.15297, October 2025; Spring Health) evaluates suicide-risk-detection performance in mental-health LLMs; handoff rate evaluates whether the system effectively bridges the user to a human resource. Both measure different things and a regime-compliant deployment should report both. Stacey et al.'s 2024 Cochrane review of patient decision aids (CD001431, sixth update) provides validated outcome instruments for related constructs (knowledge gain, decisional conflict, value-congruence) that future articulation-quality studies (§10.3) could adapt. The Patient Activation Measure (Hibbard et al. 2004; CMS MIPS 2024) is the closest established patient-side activation instrument. PAM is a longitudinal trait-level scale (per Appendix B.3); the relationship between within-session handoff and PAM trait change is at most a longitudinal-design hypothesis, not a within-session-to-PAM-stage mapping.

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7 · Limitations, anticipated objections, and population-harm risks

7.1 What this paper does not claim

This paper proposes a regime, finds prior-art evidence that the regime is currently unclaimed, and points at one reference implementation at the architectural and agency-function level only (§5). The paper does not claim that the regime obviates clinical evaluation; that articulation reliably generalizes across cultural, linguistic, and resource contexts without re-grounding; that handoff rate has been validated against clinical outcomes; that any external operator has yet adopted the regime; or that the prior-art finding rules out close-related work the author missed.

7.2 Single-deployment and single-author limitations

The reference implementation is the author's, and the analysis of the implementation is by the author. A second-source clinician audit is committed to under Wu 2026's deployment-pause posture, with a 2026-06-30 deadline. As of paper submission no licensed clinician has formally audited the reference implementation against the articulation-function claim made here. Trajectory-level evidence is deferred to future work conducted under IRB review (§10.3).

7.3 Geographic, linguistic, and care-system scope

The reference implementation operates in U.S. English (with partial Chinese coverage in safety-critical paths) and against the U.S. mental-health resource landscape. Modality vocabulary, credential structure (LCSW / LMFT / LPC etc. are U.S.-specific), level-of-care names (PHP / IOP / ASAM levels are U.S.-specific), and crisis-line numbers are U.S.-specific. Re-implementations against the NHS, Canadian provincial systems, EU national systems, or Mainland Chinese mental-health resources would test whether the regime generalizes or whether its current scope is U.S.-specific. The §A.3 search includes UK / EU patient-side AI literature; gaps remain.

7.4 Population-harm risks and structural limits

The §3.4 strata enumerate where articulation is contraindicated; this section names harm pathways at the *category* level. The acute case is covered by the deployment-pause posture (Wu 2026 §2.2) and the imminent-risk SOP (LICENSE-PROMPT §1.a); SMI, low-literacy, and cross-cultural cases are documented gaps.

The IPV lethality-vocabulary gap is below standard of care. The reference implementation does not include lethality-risk vocabulary (firearms-in-home, prior strangulation, escalation indicators per Campbell's Danger Assessment 2009) in its IPV-acuity SOP. Non-fatal strangulation is the strongest single predictor of subsequent IPV homicide (Glass et al. 2008 *J Emerg Med* 35:329–335: OR 6.70 attempted, 7.48 completed). The current SOP routes IPV signal to the DV hotline at first detection rather than performing extended articulation; lethality-vocabulary integration is a checkpoint for the §5 audit and corresponding SOP update.

Vocabulary-bounding produces structural exclusion, not edge-case oversight. §3.3 gates vocabulary on three institutional structures (referral directories, credentialing bodies, payor billing). Star (1991) predicts that any closed institutional vocabulary reproduces the IPV-class gap for populations whose distress register is not in the institutional lexicon. The §3.4 risk list is the structural output of the §3.3 choice, not a list to fix one item at a time: SMI scope-recognition, low-literacy presentations, non-Western somatic explanatory models share a root cause (vocabulary boundedness toward institutional categories). The class is the cost of

vocabulary boundedness; the regime accepts it on the grounds that bounded vocabulary is what makes the patient-portable artifact deployable at any door.

A clinician-side externality. When a user arrives at intake having been articulated by Stay, the clinician engages AI-mediated medical vocabulary in place of the lifeworld report. The Mishler (1984) tradition reads this as voice-of-medicine suppression performed *upstream* of the clinical encounter: the visible cost is reframing labor; the invisible cost is the lifeworld content that does not reach the clinical conversation at all. The §10.3 articulation-quality measurement should engage clinician-side outcomes (alliance, intake friction, diagnostic accuracy of subsequent assessment) alongside patient-side ones.

Cultural fit beyond Western individualism. Property 3 presupposes individual patient agency in help-seeking. In Confucian-influenced and other collectivist contexts, family-mediated articulation (a parent, spouse, or eldest sibling holding help-seeking authority) is the modal pattern (Kleinman 1995; Chentsova-Dutton on culturally-variant emotion expression; Karasz et al. on South Asian shame-honor; Boyd-Franklin on African-American religious-coping; Meyer 2003 LGBTQ minority-stress; Indigenous historical-trauma frameworks). The regime category travels less cleanly than the U.S. §3.3 operationalization implies; structured engagement deferred to §10.2.

Re-traumatization through phenomenological labeling. For a trauma survivor for whom "flashback" is novel and clinical, the labeling moment can be re-traumatizing or destabilizing; the trauma-informed-care literature describes the "naming brings on the experience" phenomenon clinicians manage carefully in early trauma work. §3.3 treats labeling as uncomplicatedly helpful; TIC is more cautious. A regime instance should pace labeling against signaled familiarity and treat first-introduction of a trauma label as a TIC-style choice-and-predictability moment.

Articulation under acute substance impairment. A user articulating while intoxicated produces a transcript whose authorship is compromised; the §8.4 data-ethics framing covers consent for citation but not articulation-while-impaired. The reference implementation lacks a session-quality indicator for impairment; named as a gap.

Mandatory-reporting-equivalent obligations. Articulation surfaces (child abuse, elder/dependent-adult abuse, IPV with a child in home, human-trafficking) that licensed clinicians have reporting obligations on in all 50 U.S. states. An AI is not a mandated reporter, but a deployment that surfaces these and does nothing structurally is operating in a moral grey area. The regime should specify reporter-pathway-explicit framing (route to appropriate hotline; make reporting pathway visible; do not assume user knows resources exist); the reference implementation routes to hotlines but does not document this move explicitly. Audit-flagged.

Boundary-violation patterns. Sexual-content disclosure, parasocial intensification, idealization-devaluation in BPD-presentation users, splitting between AI and human supports: repertoire issues a deployed mental-health navigation AI must have a position on. Not addressed by §3 properties or §6 metric. The leverage-prevention rule (LICENSE-PROMPT §1.b) is a partial defense (no third-party characterization to build emotional leverage) but does not cover the full pattern set.

7.5 Anticipated objections

A position paper proposing a new regime should engage the strongest counterarguments in advance. Four objections, with responses.

7.5.1 Liability laundering

The objection: "By design-refusing provider recommendation you push the failure mode onto the user; you keep the engagement value of the conversation while disclaiming responsibility for what the user does next." The regime accepts liability for the quality of the articulation (whether the modality, credential, level-of-care vocabulary supplied to the user is correct and appropriate to the presentation); this is what the §6 appropriateness audit is for. The regime does not absorb responsibility for the quality of providers the user subsequently selects, because that work is performed by existing referral systems with their own quality-control infrastructure. The split is principled, not laundering: the AI is responsible for what it outputs; the existing referral system is responsible for who it lists. Appendix B.2 names the further cost the regime accepts here (voice-of-medicine suppression performed upstream of the clinical encounter), which the directory-quality split does not address but which the regime does not pretend to.

7.5.2 Unsupervised triage by another name

The objection: "Telling a user 'this sounds like the kind of thing CBT is built for' is implicit symptom-to-diagnosis triage; the regime's claim to not do diagnosis is a paper distinction." The §3.3 three-tier distinction draws the line at fit-claims about the user's condition versus naming candidate frames. "DBT was developed for emotion-regulation difficulty" is a statement about DBT, not about the user. "You have BPD" is a statement about the user. The line is operationally fuzzy at the margin and the appropriateness audit (§6.2) is the empirical check that the line is being walked.

7.5.3 The "unclaimed regime" framing is convenient

The objection: Limbic Access with a fork-and-decouple change becomes the sixth regime, so "currently unclaimed" is a calendar fact, not a regime-defining property. The response, expanded from §3 lead-in: closest-neighbor players have publicly-positioned business arrangements that make adopting the three-property conjunction non-trivial. Several explanations are jointly plausible: switching cost from system-coupled to patient-portable output, NHS / payer audit-trail expectations that pull toward closed pipelines, brand and clinical-validation positioning that does not advertise unfiltered front-door entry, and revenue-model fit (the regime is not engagement-monetizable, §6 and §8.5). The author has not inspected contracts and the §3 claim is not "structurally infeasible" but "commercially non-aligned given current arrangements." §4.2's note that a Headspace Ebb decoupling falls into this falsifier class is consistent with this reading: the change is technically available, commercially non-aligned. The non-alignment argument is itself defeasible: a closest-neighbor with a strategic shift toward patient-owned literacy would falsify the prediction, and this paper would treat that as confirmation that the regime is real and adoptable, not as refutation. At 2026-05-02 the conjunction is unclaimed, the unclaim has multiple consistent explanations rather than one privileged mechanism, and a non-commercial deployment under donor or grant funding is the deployment shape best-positioned to ship it first because it is the shape least exposed to the commercial-alignment pressures above. A refutation would require either (a) one of these players publicly committing to portable patient-owned output, or (b) a previously unidentified system already shipping the three-property conjunction.

7.5.4 The in-scope population is not the §1.1 volume population

The objection: §7.4 excludes low-literacy, non-Western-somatic, SMI, severely cognitively impaired, minors, and acute-crisis users. §8.3 says the regime is needed where "human navigation is scarce and AI articulation is the only option." The intersection (adults with internet, literacy, English fluency, distress sufficient to seek help, no current clinician, in a region where human navigators are scarce) is a narrow subset of the population the §1.1 OpenAI flag-rate figure spans. The volume-justification population and the in-scope population are different populations. Acknowledged. The §1.1 figure is a directional signal that mental-health-relevant conversations on general LLMs are non-zero; the regime addresses a sub-population of that demand. Sizing the in-scope subset directly is future work (§10.2) and would benefit from NSDUH-grade estimation against the §3.3 vocabulary literacy band; pending that, the regime's claim is that the subset exists and is materially served by no current deployment, not a quantitative claim about its absolute size.

7.6 Regulatory fit (US state-law landscape 2025-2026)

Five 2025–2026 state statutes (Illinois HB 1806, Nevada AB 406, Utah HB 452, California SB 243, New York's AI companion safeguards) restrict AI mental-health deployment along axes the regime's three properties prima facie respect more naturally than the closed-clinical or open-companion alternatives, because the regime explicitly refuses provider recommendation and refuses fit-claims about the user's condition. The prima-facie alignment is a *deployment hypothesis* pending formal legal review by counsel licensed in the relevant jurisdictions; statutory carve-out for "patient-side literacy AI" would clarify deployability for the regime as a category. Detailed statute-by-statute analysis is deferred.

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8 · Ethics, broader impact, and conflict-of-interest disclosure

8.1 Conflict of interest

The author developed and operates the reference implementation discussed in §5 (thestay.app). The author has no commercial revenue from the implementation (it is free, MIT-licensed for code, restricted-use-licensed for the system prompt's safety-critical sections per Wu 2026) and no employment relationship with any named comparator system (Limbic, Wysa, Spring Health, Lyra, IESO, WithTherapy, Headway, Alma) or with any model-provider operating systems cited as venues of the unmet need (Anthropic, OpenAI, Google). The author has unrelated commercial projects (real-estate analytics SaaS; voice-AI tooling), none of which operate in mental health, clinical AI, AI safety governance, or any adjacent domain. The author intends to seek non-commercial donor and grant funding for continued operation of the implementation. The author has not received any institutional, foundation, or commercial funding for this paper or the reference implementation as of submission.

8.2 Broader impact

If the regime is adopted, the most likely positive impact is that some fraction of users currently

routing mental-health-relevant conversations to ChatGPT/Claude/general-purpose-LLMs route instead to a system *designed* for the use case, with safety-section retention on fork (Wu 2026), explicit refusal to substitute for clinical care, and the success metric of handoff rather than retention. The most likely negative impact is that the regime is implemented poorly by an actor without the governance scaffolding of Wu 2026; the failure mode is the NEDA Tessa case repeated. The mitigation the author proposes is the combined publication of the regime definition (this paper) and the governance pattern (Wu 2026) under restricted-use license such that derived implementations cannot strip the safety-critical sections without explicitly naming a clinical reviewer-of-record.

8.3 Equity considerations

The articulation regime presupposes a baseline of literacy, language fluency, and cognitive engagement that is not universal. Populations excluded by these assumptions (§7.4) are exactly the populations that human navigators in well-funded systems serve well; in a population where human navigation is scarce and AI articulation is the only option, the regime's deployment must include explicit fallback to "this conversation should be with a human" with the appropriate human resource named.

8.4 Data ethics

The reference implementation's session storage is local-only, end-to-end encrypted with a browser-held key, and not transmitted off the user's device (architecture summary in Wu 2026 §3.1). The diary log function is default-off and opt-in. Telemetry is metadata-only (event names and boolean flags); no message content leaves the device. The paper presents no session-level data (§5).

A trade-off for completeness: Wu 2026 §2.3's dead-man-switch heartbeat fails closed on operator unavailability, which means the bridge-with-companion-during-call function (and the rest of the deployment) returns a referral-only response when heartbeat is `expired`. This is a deliberate Wu 2026 design choice with a known clinical-side cost: a user reaching the system at 3 AM during an operator-unavailability window receives a referral, not the articulation or companion-during-call function. The trade-off is fail-closed-on-author-unavailability over fail-open-with-uncalibrated-system. The cost is named here.

8.4a Deployment posture under acknowledged gaps

The reference implementation operates as a public adult-self-attest (18+) preview under two standing conditions while the named gaps (the §7.4 IPV-acuity gap, the §4.4(iv) base-model-layer pathways, the §3.4 SMI scope-recognition gap) remain open. *Risk-band restriction*: the deployment does not advertise itself as ready for users at the IPV-acuity tier or the active-SI-with-plan tier (C-SSRS 4–5); when acuity-recognition fires for these tiers, it routes to DV hotline / 988 at the first signal rather than performing extended articulation. *Incident-log review*: operator-side signals (deployment telemetry rule-fire counts, user-reported safety issues at hello@thestay.app, external escalations) are reviewed weekly, and any incident matching the §4.4(iv) base-model-layer failure-mode shape triggers immediate pause pending clinical review. The substantive pause-trigger is the §5 clinician audit.

8.5 Funding model and sustainability

The articulation regime is structurally incompatible with engagement-monetized consumer-AI funding: a system whose success metric is handoff (§6) cannot be funded by ad inventory, subscription retention, or session-time monetization. The intended posture is donor-and-grant-supported operation under a U.S. 501(c)(3) structure (formation in progress). The Crisis Text Line / Loris.ai episode (Politico 2022; JMIR 2025) is binding cautionary precedent: conversation data from a mental-health service cannot be commercialized without breaking the trust that makes the service usable. Cost economics, candidate funding pathways, and structural-analog precedents (Signal Foundation) are documented in the project's operational materials and outside this paper's scope.

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9 · Conclusion

The bifurcation of the AI mental-health landscape into closed-clinical and open-companion has been the field's organizing framing through the early 2020s. The framing is incomplete. A user who is distressed enough to recognize that something is wrong, who does not yet know what kind of help they need, and who currently routes to ChatGPT or to nothing, defines a need that the five existing regimes (clinical, companion, crisis-line, symptom-checker, directory) do not collectively serve. *Articulation skill* names this gap as a regime, defines it by three required properties plus a scope-recognition constraint, distinguishes it from seven closest neighbors, and points at one open-source partial reference implementation at the architectural level only.

Wu 2026's governance pattern provides forkable safety scaffolding for any team building in the regime, conditional on Wu 2026's own peer-review trajectory. The primary contribution is the joint construction of two coupled elements: the §3 three-property regime (plus §3.4 scope-recognition and §6.2 defense-of-record requirement), testable against any candidate mental-health AI system; and the §6 anti-engagement criterion category (anti-engagement-under-adversary plus welfare-monotone), portable to any public-good consumer AI and operationalized within the mental-health AI vertical via the regime. Handoff rate is offered as one further operationalization of the category, scoped to system-prompt-only deployments per §6.2 and awaiting Phase-1 validation. The §4 / §A.2 prior-art finding (no current production system ships the three-property conjunction) is a *supporting empirical finding*; §7.5.3 reads it as a structural prediction about commercial incentive, not a calendar-fact gap-claim. The regime is positioned to feed users into existing referral systems, not to substitute for or compete with them. The closing claim: the three-property conjunction plus scope-recognition and defense-of-record constraints defines a regime not currently served by deployed alternatives; documented prior-art lineage exists but no patient-facing AI deployment ships the conjunction; one partial reference implementation exists with documented gaps (IPV lethality vocabulary; SMI scope-recognition; defense-of-record selection for the §4.4(iv) base-model layer; no clinician audit yet). Regulatory fit is a deployment hypothesis pending legal review (§7.6).

The regime *definition* (the §3 properties, §3.4 constraint, §4 prior-art finding, §6 metric category) stands independent of the §5 clinician audit and the Phase-1 audit. What is contingent on those events is whether the *reference implementation* counts as a regime instance once the SMI / low-literacy / cross-cultural scope-recognition gaps are or are not closed. The position-paper contribution does not stand or fall on the audit; the reference-implementation-as-regime-instance claim does.

The deployment operates as a scope-restricted public preview under standing conditions named

in §8.4a.

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10 · Future work

10.1 Clinical adoption study

The handoff-rate / handoff-appropriateness pair, audited per §5, would give a per-session clinical validity check that articulation outputs are in fact actionable by the user's downstream clinician.

10.2 Cross-cultural and cross-system grounding

Re-implementing the articulation regime against NHS, Canadian provincial systems, EU national systems, and Mainland Chinese mental-health resources would test whether the regime generalizes or whether its current scope is U.S.-specific.

10.3 Articulation-quality measurement

A small-N clinician-rated quality study against deidentified transcripts (with appropriate consent and IRB review) would be a feasible first step toward direct measurement of articulation quality beyond the handoff-rate proxy. Minimum design parameters: $n \approx 30$ deidentified transcripts; two licensed-clinician raters per transcript with crisis-intervention experience; rubric dimensions covering modality-fit, credential-fit, level-of-care-fit, vocabulary appropriateness, and downstream-actionability for a hypothetical receiving clinician; success criterion at the rubric level (Cohen's $\kappa \geq 0.6$ on each dimension as a feasibility floor, with revision if not met). Triangulation against the Cochrane CD001431 instrument family (Decisional Conflict Scale, SURE, Preparation for Decision Making) and PAM longitudinal design are research-positioning items for revision.

Service-vs-research distinction. The current deployment operates as a service (no research-question-driven enrollment, no protocol-mandated data collection, no analyses committed to publication from operational sessions) and is licensed and operated under that posture. The §6.3 Phase-1 formative audit and the §10.3 articulation-quality study above are *research* activities under 45 CFR 46.102(l) (systematic investigation designed to develop generalizable knowledge); both will be conducted under prospective IRB or independent ethics review (institutional IRB if conducted with an academic partner; WIRB-Copernicus or Advarra if conducted independently). The current operational pause-posture (§8.4a) is not a substitute for IRB review of the eventual research activities and is not portrayed as one; it is the operational standing under which the service runs while the research is not yet conducted. A reader who reads §10.3 and §6.3 as already-conducted research has misread the timeline; both are pending IRB attachment.

10.4 Operational backlog

Operational gaps in the reference implementation — harm-class items named in §4.4(iv) and §7.4, measurement gaps named in §6.1, plus deployment-scaling and research-positioning items — are tracked in the project's operational documentation and are outside the scope of this paper. The §5 clinician audit is the next external checkpoint at which harm-class status is

reviewed.

Appendix A · Methods

A.1 Reference-implementation provenance

Stay (thestay.app) was developed by the author beginning in early 2026. The system prompt, governance pattern, behavioral test suite, and reference implementation are open-source at github.com/wudaming00/stay. The engineering-ethics governance pattern is documented in Wu 2026 (Zenodo DOI 10.5281/zenodo.19941457).

A.2 Reproducibility — prior-art search log

The prior-art search referenced throughout §4 was conducted on 2026-05-02 across Google Scholar, Semantic Scholar, arXiv, PubMed, ACM Digital Library (CHI, CSCW, IUI, FAccT), and product positioning announcements. A parallel-sweep methodology covered seven research veins (HCI/CSCW venues, EU/UK regulatory frameworks, the industry scan, adjacent-field patient-navigator canon, theoretical foundations, failure-mode and regulatory landscape, and donor-funded sustainability precedents). The 14 base search queries: "AI mental health triage," "chatbot mental health navigation," "AI mental health pathway," "AI care navigator mental health," "digital mental health stepped care," "AI symptom-to-modality matching," "AI mental health literacy," "chatbot mental health intake," "AI psychoeducation," "patient navigation AI mental health," "mental health treatment matching AI," "self-referral mental health AI," "AI patient empowerment health literacy," "AI care navigation patient-side." Top 20 results per query inspected; products named (Limbic Access, Wysa Gateway/DRA, Spring Health Guide, Lyra Polaris, IESO, WithTherapy, Headway, Alma, Therabot, Babylon, Ada, K Health, Headspace Ebb, SonderMind, Talkspace, Jimini, UHC Avery, ChatGPT Health) had most recent positioning retrieved. Raw search log will be deposited as supplementary material on Zenodo. Known limitations: search did not extend to non-English literature systematically; commercial product positioning is a moving target and the verdict is current as of 2026-05-02. A systematic search of Cochrane CD001431's included studies and the IPDAS Collaboration inventory is planned for revision.

A.3 §3.3 Tier (a) vocabulary table (full)

Full representative-vocabulary table for the five sub-categories summarized in §3.3 Tier (a):

Sub-category	Representative vocabulary
Phenomenological labels	panic attack, intrusive thought, flashback, dissociation, rumination, racing thoughts, decreased need for sleep, increased goal-directed activity, hearing voices, ideas-of-reference
Modality (CBT family)	CBT, DBT, ACT, CBASP, CFT, RO-DBT, behavioral activation, CBT-I
Modality (trauma)	EMDR, CPT, PE, TF-CBT, NET, somatic experiencing, IFS
Modality (relational/dynamic)	IPT, MBT, TFP, ISTDP, AEDP, supportive-expressive, classical psychodynamic

Modality (mindfulness/ condition-specific)	MBSR, MBCT, ERP, FBT/Maudsley (adolescent ED), PCIT, prolonged grief therapy
Modality (modality-format)	group / individual / couples / family; medication management
Credential	psychologist (PhD/PsyD), psychiatrist (MD/DO), LCSW/LICSW (clinical) vs LMSW (master's, often non-billing), LMFT, LPC/LMHC/LPCC, PMHNP, prescribing psychologist (where authorized), CADC/CAC (substance use), peer support specialist, art/music/dance therapists (ATR-BC, MT-BC, BC-DMT)
Level of care (MH)	outpatient, collaborative care (CoCM, CMS CPT 99492-99494), IOP, PHP (including ED-specific PHP), RTC, inpatient, crisis stabilization, ACT (for SMI), Coordinated Specialty Care (CSC) for first-episode psychosis
Level of care (SUD)	ASAM levels 0.5–4 (4th Edition, 2023), detox, residential, sober living (Oxford House vs NARR Level III), MAT
Specialty	trauma, eating disorders, perinatal (incl. infant-parent dyadic), OCD, addiction, gambling/behavioral addictions, child/adolescent, geriatric, neurodevelopmental assessment, SMI / first-episode psychosis, personality disorder specialty, chronic pain / somatic symptom, forensic, refugee/torture, religious/spiritually integrated, neurodiversity-affirming, LGBTQ-affirming, culturally-matched

Inclusion criterion (U.S. operationalization). A term is included if it satisfies at least one of: (i) it appears in the searchable taxonomy of two or more major U.S. patient-facing referral directories (Psychology Today, Headway, Alma, Zocdoc), so a user receiving the term can find providers indexing on it; (ii) it is the canonical name used by a recognized credentialing or training organization (APA, ABCT, ISTSS, EMDRIA, ISITDP, IFI, AAMFT, ISSTD); (iii) it is named in CMS HCPCS/CPT billable-service categories or ASAM Criteria 4th Edition (2023). The three-gate choice is regime-coherent: directories, credentialing organizations, and payor billing taxonomies are the three institutional surfaces a *patient* encounters when navigating to care. Alternative gates (DSM categories, clinical-practice guidelines, treatment manuals) operate behind the patient-facing layer; vocabulary admitted by those gates but not the three above is clinician-facing rather than patient-navigatable, and including it would re-introduce the §3.3 tier-(b) line under a different name. Coherence Therapy and Schema Therapy are present in some directories but did not satisfy the two-directory threshold at the search date and are excluded from spontaneous naming (they may be discussed if the user introduces them). The reference implementation's spontaneous-naming list is a defended subset of this criterion; per-term inclusion-test mapping will be supplementary on Zenodo.

Country-agnostic form. A term is included if it satisfies (i') two or more major patient-facing referral gateways within the target jurisdiction, OR (ii') a recognized credentialing or training organization in the jurisdiction, OR (iii') the public payor's billing taxonomy. Re-implementations against UK Talking Therapies, Canadian provincial systems, EU national systems, and Mainland Chinese MH resources would substitute the jurisdiction-specific lists. The vocabulary table above is the U.S. instance of the country-agnostic three-test criterion.

Notes on the table. *(i) Phenomenological-vs-DSM disambiguation.* Tier (a) phenomenological labels are *descriptive* symptoms (what happened in the person's experience). Items that look

phenomenological but are DSM-defined episode types or forensic-grade subtypes are excluded from tier (a) and treated as tier (b). *Hypomanic episode* is a DSM-5-TR-defined episode type (4-day duration plus 3-of-7 symptoms) and the diagnostic anchor of Bipolar II. *Command hallucination* is a forensic-grade subtype with specific risk-assessment protocols around compliance, content-congruence with delusions, and harm-target (Hersh & Borum 1998 *JAAPL* 26(3):353–359; McNiel, Eisner, & Binder 2000 *Psychiatric Services* 51(10):1288–1292). The system may name "racing thoughts," "decreased need for sleep," "elevated mood," and "hearing voices" descriptively; it does not name "hypomanic episode" or "command hallucination," because doing so collapses to a tier-(b) diagnostic statement with extra steps. *(ii) Candidate-modality deferrals.* The expanded modality list includes additions (MBT, TFP, FBT, CBT-I, behavioral activation, CFT, NET, PCIT, prolonged grief therapy) listed pending per-term verification against the three inclusion gates; per-term gate-by-gate mapping is committed as supplementary material on Zenodo before deployment-spontaneous-naming inclusion. Four candidates are deferred pending §3.4 stratum specification: PCIT (parent-child therapy serves children, §3.4 scopes out minors); FBT/Maudsley (canonical adolescent ED modality, family-based child therapy by definition); MBT and TFP (BPD evidence-base modalities, but §3.4 is silent on PD-specific scope and §7.4's boundary-violation paragraph names BPD-presentation users as a regime-level gap). *(iii) Credential disambiguation.* LCSW/LICSW indicates clinical licensure with independent psychotherapy billing; LMSW is a master's social-work credential that often cannot independently bill psychotherapy. CADC/CAC is the dominant SUD-treatment workforce in residential and IOP settings. LPC/LMHC/LPCC are the same role under different state-naming conventions. LMFT in some states (e.g., California) is functionally equivalent for relational/family work. State scope-of-practice variance is acknowledged as a deployment-variance issue. *(iv) MH and SUD level-of-care continua* do not map onto each other; an articulation system that knows only the MH continuum will mis-route SUD presentations. *(v) ACT disambiguation.* "ACT" appears twice with different referents: in the modality row = Acceptance and Commitment Therapy (Hayes); in the level-of-care row = Assertive Community Treatment (Stein & Test, 1980). The system disambiguates by context. *(vi) LOC acronym glossary.* CoCM = Collaborative Care Model (CMS CPT 99492-99494); IOP = Intensive Outpatient; PHP = Partial Hospitalization Program; RTC = Residential Treatment Center; CSC = Coordinated Specialty Care (first-episode psychosis); MAT = Medication-Assisted Treatment; ASAM = American Society of Addiction Medicine; NARR = National Alliance for Recovery Residences.

Appendix B · Lineage disambiguation (pointer)

§1.4 names five literatures the regime engages. Full per-literature disambiguation across six subsections (B.1 Strauss/Star and the boundary-object disambiguation; B.2 Mishler/Kleinman and the harm-reduction-relative-to-counterfactual frame; B.3 Nutbeam/Hibbard and the within-session-vs-longitudinal measurement gap; B.4 Elwyn/decision-aids and the three structural differences; B.5 CMS PIN function-class kinship; B.6 AI-safety canon partitioned by deployment-layer vs training-layer engagement) is published as supplementary material accompanying this preprint on Zenodo. Body sections that depend on Appendix B engagement (the §1.4 lineage list, §3.3 vocabulary-bounding rationale, §4.4(iv) base-model-substrate inheritance, §6.2 Goodhart engagement, §7.4 risk-population structural-output finding, §10.3 articulation-quality measurement) name what they depend on inline.

Supplementary: Appendix B

Appendix B: Lineage disambiguation (supplementary)

Companion supplementary to *Articulation Skill: A Patient-Navigator Regime for Public-Good Mental-Health AI* (Wu 2026, Zenodo DOI 10.5281/zenodo.20028720). The main paper's §1.4 names the five literatures in summary; this file does the full disambiguation.

Appendix B · Lineage disambiguation (full)

§1.4 names five literatures the regime engages. The disambiguation pattern is the same in each section: name the borrowed concept, identify the structural difference that prevents the regime from being read as a sub-type, and mark the predictable cost the borrowed tradition's own critics would name. The map across sections is: B.1 (Strauss/Star) sets the patient-side-capability framing the rest depends on; B.2 (Mishler/Kleinman) is the inverse-direction case where the regime moves *toward* the institutional voice the tradition critiqued; B.3 (Nutbeam/Hibbard) is the measurement-tradition kin; B.4 (Elwyn/decision-aids) is the closest sibling, distinguished by front-door context (B.4 (c)); B.5 (CMS PIN) is function-class kinship not billing-pathway candidacy; B.6 (AI-safety) partitions citations by deployment-layer engagement vs training-layer residual.

B.1 Strauss articulation-work, Star category-work, and the artifact's genre

Strauss (1985, 1988, 1993) defined "articulation work" as cooperative coordination labor across multiple actors with different roles — the invisible work that meshes fragmented care delivery. Schmidt and Bannon (1992, *CSCW: An International Journal* vol 1) extended this with the claim that articulation work is required precisely because complete standardization is impossible. Our "articulation skill" names a unilateral patient-side capability of translating undifferentiated experience into a bounded institutional vocabulary; this is closer to Star (1991) on category-work and the standardization-of-conventions cost — the labor of matching one's experience to institutional categories not built to fit (Star, "Power, technology and the phenomenology of conventions: On being allergic to onions," in J. Law (ed.), *A Sociology of Monsters*, Routledge; and Star (1991), "The sociology of the invisible: The primacy of work in the writings of Anselm Strauss," in D. Maines (ed.), *Social Organization and Social Process*, Aldine de Gruyter) — than to Strauss articulation work in the cooperative CSCW sense. The term "articulation skill" is repurposed: it names the patient-side capability of "finding the words for it," not the cooperative-coordination labor across distributed actors that Strauss and Schmidt-Bannon defined. Strauss articulation work is constitutively cooperative; the regime's articulation skill is constitutively unilateral (one user, one AI, no role-coordination). A reader expecting the cooperative meaning will find the regime does not deliver it. The repurposed term is closer to Star's (1991) category-work in operational shape; we use "articulation" because that is the

colloquial English term for what users do when they translate experience into clinical-adjacent vocabulary, and the colloquial use is what the regime directly addresses. The term invokes a Strauss/Schmidt-Bannon tradition whose canonical move (filling gaps that standardization cannot reach) is in tension with the §3.3 design choice (binding vocabulary to institutional standards). The §3.3 inclusion criterion places the regime on the *standardization* side of the Schmidt-Bannon distinction, and the §7.4 risk-population list is the predictable Star-tradition output of that choice, not a list of items to fix one at a time. The two traditions predict different things about the regime: Schmidt-Bannon predicts that cooperative articulation work will still be required downstream (clinician and patient renegotiating what the artifact misses), and the regime accepts this — the artifact is preparation for that conversation, not a substitute for it. Star predicts that the standardization itself imposes costs on those who do not fit the categories, and §7.4 is that cost made explicit. Bounded vocabulary is what makes the patient-portable artifact deployable at any door; that is the choice made, and it is not a choice Star's tradition would endorse.

On the term itself: a CSCW reader will reasonably read the choice to call this regime "articulation skill" — given that "articulation work" is a term of art with 40+ years of cooperative-coordination meaning — as disciplinary appropriation of a term whose canonical meaning is the inverse on both axes (cooperation vs unilateral, gap-filling vs standardization-binding). Three reasons we keep the term. (i) "Articulation" has substantial colloquial use in clinical and intake settings predating and parallel to the CSCW term-of-art ("the patient articulated their concern as…", "help the patient articulate what they are experiencing"). The colloquial usage is the source the regime addresses; the CSCW lineage is named to disambiguate, not to claim. (ii) The regime sits in continuity with Strauss-Schmidt-Bannon at the level of the ecosystem the artifact participates in: Schmidt-Bannon's prediction (cooperative articulation work is still required downstream of any artifact) is one of the regime's load-bearing acceptances — the artifact is preparation for that downstream work, not a substitute. The terminological link signals this continuity even where the design choice on the standardization axis inverts. (iii) The naming alternatives we considered (patient-side translation skill, intake preparation skill, navigation literacy) each lose a different piece of what the regime does; a name is judged against the alternatives that exist, not against an ideal name that would not invoke any tradition.

The §3.1 patient-portable literacy artifact is, as noted above, a *delegated artifact* or portable referral letter rather than a boundary object. The CSCW genre literature on coordinative paper-form artifacts (Berg & Bowker 1997 *The Sociological Quarterly* on the medical record as a coordinative artifact; Sellen & Harper 2002 *The Myth of the Paperless Office*) predicts specific failure modes for artifacts in this genre — selective transcription, format-driven omission, the tendency of "what the next reader needs" to crowd out "what the patient said," and reciprocal pressure on the next reader to treat the artifact as authoritative rather than provisional. The §3.1 artifact is structurally exposed to these; §10.3's articulation-quality study is where the regime's actual exposure to them would be empirically tested. The genre literature is sharper than the boundary-object frame and is the tradition the §10.3 measurement work should engage in revision.

The §3.1 literacy artifact is not a boundary object. A boundary object (Star & Griesemer 1989; Star & Bowker 1999) requires plasticity to local needs of several parties, robustness across translations, and sitting *between* communities of practice. The §3.1 artifact is produced by one community (AI + user) and read by another (downstream clinician); it is a *delegated artifact* or portable referral letter.

B.2 Mishler voice-of-medicine and Kleinman explanatory models

Mishler (1984, *The Discourse of Medicine*) argues that the clinical encounter is a power-asymmetric translation in which voice-of-medicine *suppresses* voice-of-lifeworld. Kleinman (1978, 1980, 1988 *The Illness Narratives*, 1995 *Writing at the Margin*) argues that patient explanatory models have clinical and ethical weight that biomedical translation strips, and that the *clinician* is supposed to negotiate the gap, not standardize toward the biomedical. The articulation regime performs upstream the standardization Mishler critiques and Kleinman asks the clinician to resist: an AI that translates the patient's lifeworld content into modality and credential vocabulary before the clinician sees the patient is doing voice-of-medicine suppression with extra steps. Mishler's voice-of-medicine is moved earlier in the clinical pathway under this regime; that is the move, and naming it as such matters more than calling it an inversion. The §3.3 vocabulary-bounding is the explanatory-model standardization Kleinman's tradition resists.

The structural critique applies: this is voice-of-medicine suppression performed upstream of the clinical encounter, eliminating the site where Kleinman asks the clinician to negotiate the patient's explanatory model against the biomedical. This is the inverse direction from B.1: there the regime accepts a *Star-tradition* cost (standardization disadvantages those who do not fit categories) as the price of portability; here the regime moves *toward* the institutional voice that Mishler and Kleinman identified as the locus of harm. The cost is non-trivial; the comparator below addresses but does not erase it.

What the regime offers instead is *harm-reduction relative to a counterfactual*. The relevant comparator is not the clinical encounter Mishler diagnosed but the current state for the in-scope subset (§7.5.4): users routing mental-health-relevant conversation to ChatGPT/Claude/general-purpose-LLMs, where (a) the same lifeworld-to-vocabulary translation occurs without explicit lineage acknowledgment, (b) the negotiation Kleinman calls for is *equally* foreclosed, (c) the failure modes documented in §4.8(i)–(iv) (sycophancy, confabulation, no-safety-section-retention-on-fork, AI-induced psychosis) are unbounded by any stated regime. The regime's claim is that *given the counterfactual*, a deployment that performs the suppression *with named lineage tensions, bounded vocabulary, scope-recognition constraint, and articulation-quality measurement as a research program (§10.3)* is harm-reductive in expectation, not harm-eliminative. A reader who treats the comparator as the well-resourced clinical encounter Mishler diagnosed will judge the regime as net-harm; a reader who treats the comparator as no-fitted-destination plus general-LLM venue will judge it as net-helpful. The in-tradition response to the first reader is deferred to §10.3, where articulation-quality measurement against downstream-clinician outcomes is the empirical pathway by which the harm-reduction claim can be tested against the tradition's standards. §7.4 names the cost.

B.3 Nutbeam health literacy and Hibbard PAM

Nutbeam (2000, *Health Promotion International*) defined a three-tier health-literacy framework (functional / interactive / critical). The §3.3 vocabulary tiers operate in the interactive-literacy band; whether the regime *builds* critical-literacy capacity or *substitutes* for it is an open empirical question we do not resolve. Hibbard et al. (2004, *Health Services Research*) developed the Patient Activation Measure (PAM) as a longitudinal trait-level instrument; PAM stages 1–4 are stages of multi-month change, not session-level states. Within-session articulation outcomes are not directly comparable to PAM stage transitions; the relationship is at most that PAM trait change might be triangulated against repeated articulation-session outcomes over months, a

longitudinal study the regime has not specified. The PAM measurement tradition and the decision-aids measurement tradition (B.4: O'Connor 1995 DCS, IPDAS) are the two closest measurement-side neighbors; §10.3's articulation-quality construct is the work neither has standing instruments for.

B.4 Elwyn shared-decision-making and patient-decision-aids — kin, not sub-type

Elwyn et al.'s shared-decision-making framework (2012) operates inside the clinical encounter; Elwyn's later work and the patient-decision-aids tradition (Stacey et al. 2024 Cochrane review CD001431, sixth update; the IPDAS Collaboration; validated instruments including the Decisional Conflict Scale (O'Connor 1995), Decision Self-Efficacy Scale, SURE test, and Preparation for Decision Making scale) sit upstream of SDM, in the same slot the articulation regime occupies. Three structural differences hold the regime claim against the decision-aid-sub-type reading:

- **(a)** The decision-aids tradition presupposes a treatment decision (Elwyn 2012; the IPDAS defining criterion is preparation-for-decision). Articulation skill operates upstream of any presupposed decision and supports users who do not yet know what kind of decision they face.
- **(b)** The decision-aids tradition is structured around evidence-based options for a defined clinical condition (CD001431 includes decision aids for screening, surgery, treatment selection given a known diagnosis). Articulation skill operates without a known diagnosis and refuses to render one (Property 3 tier b).
- **(c)** The decision-aids tradition assumes the user has already entered the clinical pipeline that selects them in (referrals, intake screens, clinician judgment that a decision is approaching); the regime operates in an *unfiltered front-door context* where no upstream pipeline has performed that selection. This requires the regime to take on a filtering layer (the §3.4 scope-recognition constraint) the decision-aids tradition does not need. The structural difference is the front-door-vs-filtered context, not "the regime has §3.4."

These are structural differences sufficient to position the regime as adjacent to and upstream of decision aids, not a sub-type within them. Kinship and distinction both hold. The §4 prior-art search will be extended in revision to inspect Cochrane CD001431's included studies and the IPDAS inventory; results in §A.2.

B.5 CMS Principal Illness Navigation — function-class kin, not billing-pathway candidate

CMS established Principal Illness Navigation billing codes (G0023, G0024) effective 2024-01-01, formalizing the patient-navigator role for serious-illness patients (cancer first, extensible to others). CMS PIN reimburses *human* labor performed by a trained team member with documented competencies, with documented warm handoff and confirmed follow-through. The articulation regime as deployed has none of these (no clinician supervision relationship, no documented competency framework, no team integration, no warm handoff with confirmed contact). The §6.1 handoff metric is documented within-session intent only, not documented follow-through. The lineage citation here is *function-class kin, not billing-pathway candidate*: the regime occupies the same agency-function neighborhood as PIN-reimbursed human navigators (translating clinical complexity, preparing patients for handoff), but it is not on a near-term

reimbursement trajectory and is not positioned as one. The relationship to B.4 is parallel: there the regime is *upstream of* the decision-aids slot (decision is not yet identified); here the regime is *upstream of* the PIN slot (the patient has not yet entered a navigator-eligible care relationship). The September 2025 *Journal of Oncology Navigation & Survivorship* AI-navigators issue and Montefiore Einstein's MyEleanor (33% rescheduling among 2,400 nonadherent colonoscopy patients, ASCO 2024) are adjacent AI-navigator deployments in oncology; the articulation regime is the mental-health-AI sibling, with three additional operational constraints (no provider recommendation, system-agnostic patient-owned output, vocabulary scope deliberately bounded). The contribution is the application and operationalization in the mental-health AI context.

B.6 AI safety / specification-gaming canon

The §6.2 gaming-mode analysis is *adversarial reward-shaping* analysis (proposing a metric and asking which games it enables). The borrowing carries the safety community's predictions about which gaming patterns are addressable at the deployment layer (system-prompt level, against a frozen base model) versus at the training layer (RLHF preference signals, base-model behavior). The regime's defenses are at the deployment layer only; training-layer references are cited to characterize residual risk the deployment layer cannot fully neutralize.

Deployment-layer relevant (the regime's defenses bite here): Krakovna et al. 2020 (specification gaming, DeepMind blog and curated-examples list) supplies the gaming-mode taxonomy §6.2 enumerates against; Goodhart 1975 with Manheim & Garrabrant 2018 (Goodhart variants, arXiv:1803.04585) supplies the metric-pressure framing the rubric-as-metric is exposed to; Perez et al. 2022 (model-written evaluations, arXiv:2212.09251) supplies the auto-generated red-teaming methodology that is the candidate operationalization for the "stratified red-teaming of the rubric" defense named in §6.2 against §6.3's recursive-Goodhart concern.

Training-layer residual (the regime's defenses do not reach these; cited as bounds on what deployment-layer work can claim): Hubinger et al. 2019 (mesa-optimization, arXiv:1906.01820) and Shah et al. 2022 (goal misgeneralization, arXiv:2210.01790) describe failure modes of the base model that deployment-layer prompting cannot detect or correct; Sharma et al. 2023 (sycophancy, arXiv:2310.13548) and Casper et al. 2023 (RLHF problems, arXiv:2307.15217) describe RLHF-trained dispositions that deployment-layer instruction may dampen but does not remove. Their presence in the citation set is to mark the residual the regime carries from the base model into deployment, not to claim the regime addresses them. The Hubinger 2019 citation deserves a sharper treatment: mesa-optimization is a property of *the base model the deployment runs on*, not only of a deployer's training pipeline. A frozen base model may already carry mesa-objectives developed during its own training that are misaligned with articulation quality on the distress distribution; the deployment-layer system prompt does not detect, measure, or correct these. The §6.2 audit's appropriateness rubric is the only deployment-side instrument with even partial bite on this concern (it can detect behavioral signatures at the conversation-arc level), and the rubric is not built specifically to detect mesa-objective signatures. We name this as a residual the deployment substrate carries, not as a problem the regime addresses; a mesa-objective-aware audit dimension is a candidate future extension, not a current capability.